

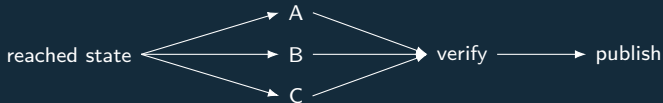
Forkable Sandboxes

**Forkable sandboxes need explicit contracts
for state, evidence, and authority.**

Yossi Eliaz

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Disclosure: I work for a company that builds sandboxes.



The branching mechanism must preserve the state the continuation needs.



- **Source alternatives:** a worktree may suffice.
- **Prepared files:** compare a warm image with a disk snapshot.
- **Running state:** verify process or VM capture. Hosted-model sampling stays external.

Proposed lifecycle; the serial factory did not establish a necessary live process state.

Existing runtimes reuse state for cloning, evaluation, and recovery.

System	Reused state and operation
SnowFlock	VM_fork(N) clones reached VM state.
Kimi AgentENV	Checkpoint, resume, and branch the sandbox for reward evaluation.
DeepSeek DSec	Retain the sandbox and replay cached command results on resumption.

Kimi reports **51.2 million sandboxes** across training and evaluation.

A sandbox branch for judging does not imply a forked judge.

SnowFlock, EuroSys 2009; Kimi K3, §5.3.2; DeepSeek-V4, §5.2.5.

A backup request reduced p99.9 latency from 1,800 ms to 74 ms.

Read 1,000 BigTable keys.

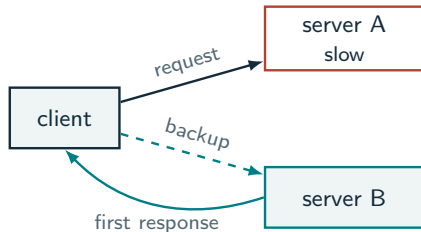
Send the backup after 10 ms. Accept the first response.

1,800 ms → **74 ms**

p99.9 latency, with 2% additional requests

Shared latency causes limit the benefit.

Dean & Barroso, *The Tail at Scale*, CACM 56(2), 2013.



250 attempts raised SWE-bench Lite coverage from 15.9% to 56%.

DeepSeek-Coder-V2-Instruct; issues solved by at least one sample.

Attempts per issue	Issue coverage
1	15.9%
250	56%

- Coverage counts whether a correct candidate exists in the collection.
- Production selection still needs reliable verification.
- False acceptance costs can favor fewer than ten attempts.

Brown et al., arXiv:2407.21787; Stroebl et al., arXiv:2411.17501.

We built evidence accounting; the integrated forked factory remains proposed.

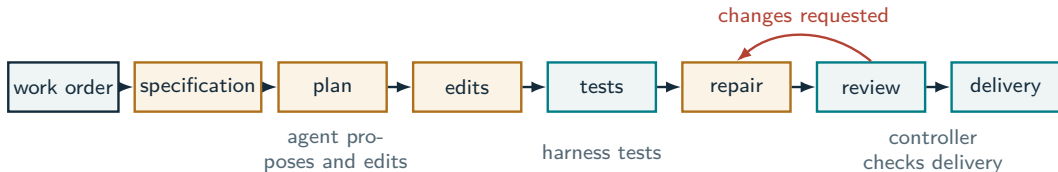
Status	Established scope
MEASURED	Two factory work orders and distinct sandbox API paths.
BUILT	Duplicate-evidence rejection, lineage transport, artifact digest gate.
PROPOSED	Integrated fork lifecycle, dependence-aware uncertainty, restore authority.

No case-study run forked.

Fork speedup and sibling dependence remain unmeasured.

SELFHOST-2/3 records; Eliaz, arXiv:2607.09689.

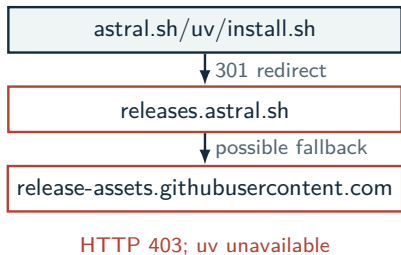
The factory ran one hosted model through separate stages and checks.



- Docker Sandboxes through Airflow's sandbox toolset.
- One hosted model through Databricks across agent roles.
- Unmanaged cell, harness-answered gates, serial execution.

SELFHOST-2, 27 September 2026; `serial_fallback_missing_fork`.

The uv installer failed at two redirect hosts missing from the policy.



Six allowed hosts; two destinations missing.

Observed on **islo**; the work order then ran on **Docker Sandboxes**.

A shared release-asset hostname grants access beyond this installer.

Installer observation: islo CLI 0.53.1, 27 September 2026; SELFHOST-2 intent.

The approved plan limited the allowlist update to six files.

Planned change	Files
Python allowlist and shell bootstrap	2
Regression tests	1
Deployment script	1
Documentation	2
Total	6

- Both hosts must appear in code, scripts, and documentation.
- Hermetic checks must assert agreement between declarations.
- Dispatch ownership lies outside this plan.

SELFHOST-2 plan and plan approval; approval answered by the harness.

Two repairs and two reviews used 59% of recorded stage time.

Stage	Work performed	Time	Cost
Specification and plan	Acceptance conditions; six-file scope	3:41	\$0.96
Build and test	Three edits; suite failure; first repair	20:01	\$4.50
Review	Review; second repair; tests; re-review	18:55	\$4.87

43:44 elapsed 42:38 recorded stages **\$10.33 model cost**

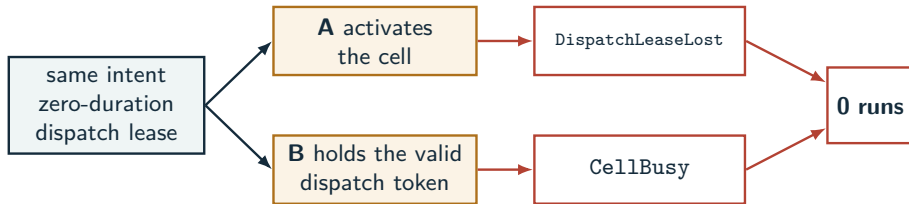
Repairs and reviews: 25.2 minutes, \$7.75. That is 59% of stage time and 75% of model cost.

Setup: **20.3 s**. The whole reusable prefix was not measured.

SELFHOST-2 stage metrics and agent-session records; costs at list price.

The repair agent diagnosed different owners for the lease and active cell.

Concurrency test: “0 runs from 4 replicas”.

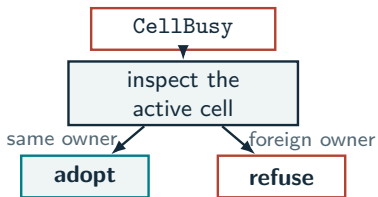


This is the agent's recorded diagnosis, without independent validation.

This dispatch-coordination lease differs from the unused sandbox authority lease.

SELFHOST-2 first repair record, agent/fix.4.json.

The suite had no failures, but the adoption path lacked a direct test.



1,972 reported cases, including skips; 0 failures.

The same-owner CellBusy path lacked a deterministic test.

Required: force that path and assert exactly one dispatch.

SELFHOST-2 repair and review records; suite run by the harness.

The second repair changed code while the requested test remained absent.

Step	Recorded decision
Review 1	Request deterministic tests as a blocker.
Repair 2	Report protected tests; refactor a recovery path onto the adoption helper.
Review 2	Approve; rate the missing direct test minor.

15 allowed edits, including 6 by repairs. The test-edit hook never fired.

No blocked test-edit attempt was recorded.

PROPOSED Explicit scope amendments and behavioral-obligation checks.

SELFHOST-2 edit-hook log, repair outputs, and reviews.

Delivery was refused three times, and cleanup lost the patch.



- Separate work order: **about 26 minutes, \$3.43, no PR.**
- The unexpected file was our harness's review archive.
- **PROPOSED** Export the candidate and evidence before teardown.

SELFHOST-3 delivery and recovery records; redacted copy on request.

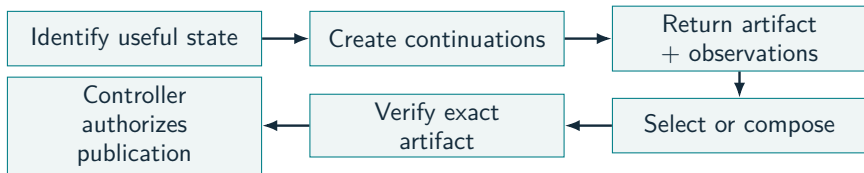
The records included skipped checks, mislabeled approvals, and missing inputs.

Recorded object	Limitation
CI success	Evaluation skipped without a key; the job passed in about 3 s.
Human approval	The harness answered, while the record named a human actor.
Adapter digest	The shim's source remained outside Git and unrecoverable.

Receipts need an execution status, actual actor, and artifact identity.

CI, 28 September 2026; SELFHOST-2/3 approvals and adapter records.

The observed gaps define obligations that future branches must preserve.



- **Observed needs:** explicit tests, accurate receipts, durable export.
- **Proposed branches:** identified parent, private continuations, fresh authority.
- This case establishes files and feedback, without a demonstrated need to preserve live processes.

SELFHOST-2/3 observations; the integrated branching lifecycle is proposed.

Worktrees, checkpoints, and replay preserve different kinds of state.

State to reuse	Tools	What a continuation receives
Source/build state	Git worktrees, OBuilder	Files and cached build inputs
Process state	CRIU, gVisor save/restore	Supported process/kernel state
VM state	Xen, SnowFlock, Firecracker	Declared memory and device state
Recorded observations	rr, DSec replay	Recorded execution or responses

Choose the mechanism for the state the continuation actually consumes.

Applying a code patch may require reload or restart.

CRIU, gVisor, Firecracker and rr documentation; SnowFlock, EuroSys 2009; DeepSeek-V4 §5.2.5.

Published latency figures measure different operations.

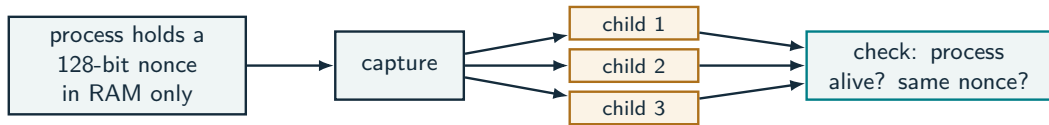
System	Operation reported	Value	Source
DeltaBox	checkpoint	10.83 ms	arXiv 2026
Kimi K3 AgentENV	checkpoint	as low as 133 ms	tech. report §5.3.2
Kimi K3 AgentENV	resume	as low as 49 ms	tech. report §5.3.2
Shepherd	fork	134–143 ms	arXiv 2026
SnowFlock	remote fork across hosts	600–800 ms	EuroSys'09
Firecracker	boot to application code	under 125 ms	NSDI'20

Each value needs its operation, workload, endpoint and timing convention. The rows cannot rank platforms.

Compare equivalent usable states before comparing latency.

[DeltaBox](#); [Kimi K3 §5.3.2](#); [Shepherd](#); Lagar-Cavilla et al., EuroSys'09; Agache et al., Firecracker, NSDI'20.

Restore fidelity needs checks of the state the API claims to preserve.



Result

Permitted description

files survive only

filesystem restore

process and RAM survive

evidence of process and memory restoration

declared semantic checks match

fidelity for those checked properties only

Passing is necessary for a memory claim. Full fidelity needs more. Compare declared outputs, ignoring timestamps. Worktrees, CRIU, VM snapshots and replay logs restore different state.

Gate 0 is preregistered and unrun, so we say restore fan-out.

Our preregistered protocol, Gate 0. [Kimi K3 §5.3.2](#). CRIU and Firecracker snapshot documentation.

Each child needs private writable state and host-controlled authority.

Inherited or shared	Required contract
Immutable parent/cache	Digest identity; declared trust domain
Writable branch state	Private to the child
Child permissions	Host authenticates and authorizes identity, epoch, expiry and budget
Publishing credential	Controller holds it; approval names the artifact

Kernel isolation does not make a copied bearer token harmless.

Firecracker and gVisor documentation; Capsicum, USENIX Security 2010; CHERI, IEEE S&P 2015.

Restoring local state does not undo completed external actions.

Effect	What restore cannot undo
Sent model request	The remote invocation and its cost
Sent tool request	A remote action that may have completed
Staged patch	Local bytes can be retained; publication needs a separate gate

- Proposed effect journal and buffer: reconcile uncertainty or use idempotency.
- Export artifacts and receipts before every teardown.

Speculator, SOSP 2005; external synchrony, OSDI 2006; Remus, NSDI 2008; DeepSeek-V4 §5.2.5.

Forking saves resources when avoided preparation exceeds reuse overhead.

$$(N - 1) P > H + N (R + D)$$

preparation P , capture H , restore R , divergence D , children N . Additive resource-cost model.

Calculation, not a measurement. Units: resource-seconds.

Case	Avoided preparation	Added reuse cost	Result
$N=8, P=30, H=2, R=1, D=3$	$7 \times 30 = 210$	$2 + 8 \times 4 = 34$	fork saves 176
same, warm cache $P=2$	$7 \times 2 = 14$	34	fork loses 20

Batch elapsed time is a separate question: it depends on critical paths and contention. A faster-booting minimal image is the alternative (Jitsu, LightVM).

Madhavapeddy et al. NSDI'15 (Jitsu); Manco et al. SOSP'17 (LightVM, NEC Labs). Parameter values are illustrative.

The API traces measured creation and restore–run–capture separately.

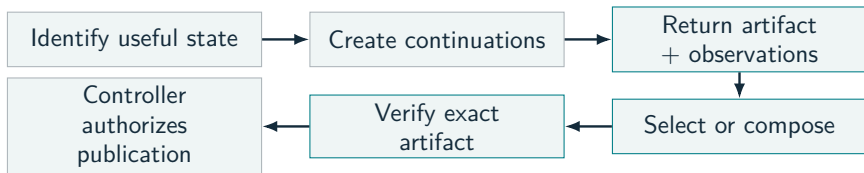
Workflow	Success	p50	p95
Daytona: create	1,024/1,024	0.20 s	1.12 s
Tensorlake: create	256/256	3.44 s	9.00 s
islo: restore, run, capture	255/256	6.87 s	9.04 s

- Restore path: a 141 MB named snapshot at concurrency 12
- Whether the snapshot captures memory is unverified, so the third row is restore fan-out
- Teardown is excluded; operations and workloads differ.

The table establishes exercised API paths and their latencies.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), Table 2.

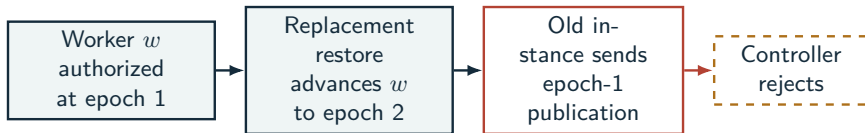
The factory has a publication gate, but its fork store is missing.



- **Exercised:** scheduler, journal, work cell and artifact-digest gate.
- **Built separately:** reducer; lease broker unused in these work orders.
- **Missing/proposed:** fork store and external evaluator.

Factory records and design; arXiv:2607.09689v4.

A replacement restore must invalidate the old worker epoch.



- Evidence, approval and publication name one frozen artifact.
- Concurrent fork gets a new worker identity; the parent may continue.
- TLA+ design exists; no model-checker run is recorded.

Proposed promotion protocol; Gray & Cheriton, Leases, SOSP 1989.

Selection, composition, and pooling require different evidence.

Returned objects	Decision	Required evidence
Different patches	Select one	Validate the selected artifact
Compatible patches	Compose	Test the exact composition
Measurements of one target	Reduce	Calibrated information and dependence model

The implemented numerical reducer occupies the measurement lane.

A patch score is not automatically a precision estimate.

Factory records; Evidence-Aware Reduction for Forkable Compute, arXiv:2607.09689v4.

The Chamo project rejected a numerically unresolved cross-check.

Ori Chamo and Yossi Eliaz: Three-Body Orbit Atlas.



- 135,445 source orbits; 26 selected difficult links continued bidirectionally.
- Apparent projected branches connected within one sampled component.
- An unresolved independent RK4 cross-check was rejected as evidence.

Chamo–Eliaz preliminary preprint; Atlas records. Completeness/release gates remain open.

Source identities must be recorded before counting separate evidence.

Visible count	Relationship to retain	Claim boundary
Cells	Patient identity	Cells may share one source
Mining addresses	Linked agents	Addresses can belong to one actor
Executions	Observation and case IDs	New execution can reuse old evidence

Record the unit of analysis and its relationships with each result.

Blackburn et al., [arXiv:2206.02871](https://arxiv.org/abs/2206.02871); Saurty-Seerunghen et al., iScience 2026; provenance semirings, PODS 2007.

Checkpoint-started success is conditional on reaching that checkpoint.

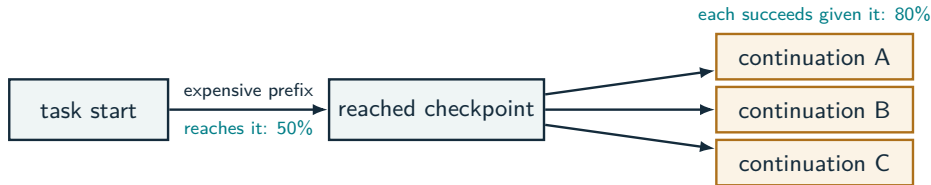


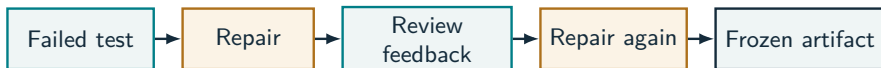
Illustration. end-to-end success = $0.5 \times 0.8 = 0.4$, while a checkpoint-started run shows 0.8.

- A checkpoint distribution differs from the task-start distribution.
- When every repair inherits one plan, continuation success alone does not validate every plan choice.

Continuation success alone does not estimate how often the checkpoint is reached.

Ecoffet et al., [First return, then explore, Nature \(2021\)](#). Probabilities are illustrative, not measured.

Repair tests and generalization tests support different claims.



- A regression check supports the deterministic behavior it asserts.
- Feedback used in development changes the candidate.
- Generalization and selection claims need an appropriate untouched evaluation.

Dwork et al., Science 2015; Russo & Zou, AISTATS 2016; Kimi K3 §4.2.6; factory records.

A check can run correctly and still miss the required behavior.

Question	Same-owner adoption case	Mechanism
Execution integrity	Did the check run on this digest?	Artifact-bound recorder
Behavioral coverage	Did it force CellBusy adoption and assert one dispatch?	Deterministic targeted test
Statistical support	What broader claim do repeated outcomes support?	Suitable sampling and dependence model

Duplicate rejection cannot detect a missing behavioral assertion.

SELFHOST-2 review records; proposed acceptance contract.

Distinct execution IDs do not make copied evidence new.

Returned evidence	Treatment
A and B both return observation e_1	Reject declared duplication
A returns e_1 , B returns e_2 ; parent p shared	Retain ancestry; dependence unresolved
100 cases run four times	400 outcomes within 100 case identities

Keep execution, observation, case and candidate identities separate.

A fresh stochastic evaluation may be new evidence while retaining its shared-source labels.

Reducer implementation, arXiv:2607.09689v4; technical versus independent replicates, Gilad et al., Commun. Biol. 2021.

At correlation 0.1, 100 measurements have the mean precision of about nine.

$$\text{Var}(\bar{Y}) = \sigma^2 \left[\rho + \frac{1-\rho}{N} \right]$$

$$N_{\text{eff}} = \frac{N}{1+(N-1)\rho} = \frac{100}{1+99(0.1)} \approx 9.17$$

- Equal marginal variances; common pairwise correlation.
- Describes variance-equivalent precision of a mean.
- Does not remove shared bias or give a probability of correctness.

Variance calculation; Kish, Survey Sampling, 1965. Values are illustrative.

Zero pairwise correlation does not fix the all-wrong probability.

Calculation. Three candidates. In each vector, 1 = candidate wrong. Listed vectors are equally likely.

Error vectors	Individual error rate	Pairwise correlation	All wrong
{000, 011, 101, 110}	50%	0	0%
all eight vectors	50%	0	12.5%
{001, 010, 100, 111}	50%	0	25%

Same marginals and zero pairwise correlation, yet the all-wrong rate ranges from 0% to 25%.

A selector that returns one candidate's answer succeeds with probability at most $1 - P(\text{all wrong})$.

An effective sample size describes the precision of an average. It is not a guarantee for search.

Search coverage depends on joint failure; selection is an additional task.

Calculation by enumeration. Ceiling: [Chen, arXiv:2606.27288 \(2026 preprint\)](#), ceiling $1 - \beta$, β the all-fail rate.

Shared conditions can reduce uncertainty in a paired comparison.

Compare A and B on the same case and controlled conditions.

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B)$$

- Positive shared variation can reduce variance of the difference.
- Record ancestry separately from shared fixtures, tests and feedback.
- Control the relevant randomness; guest seeds do not control a hosted model.

The useful sharing policy depends on the quantity being estimated.

Common-random-numbers methodology; covariance identity; proposed runtime manifest.

Passing every pair does not establish that the full composition passes.

Illustration: each patch enables one worker; capacity permits two.

Patches applied	Workers	Capacity check
Each single patch	1	✓ pass
Every pair	2	✓ pass
All three	3	✗ fail

- Eight candidates have 28 pairs and 256 subsets.
- Pairwise conflict checks can miss higher-order effects.
- Verify the exact selected composition and bind its digest.

Illustrative capacity example; composition requirement in proposed branch contract.

The reducer rejects declared duplicates and retains lineage.

Result record	Estimate, reported information, sample count, evidence IDs, lineage, execution metadata
Repeated declared evidence ID	Merge raises; duplicate is rejected
Distinct IDs with shared parent	Merge proceeds; lineage is retained
Dependence correction / refusal	Proposed; not implemented

Field validation does not establish calibration or a common target.

Reference reducer; Evidence-Aware Reduction for Forkable Compute.

Information pooling outperformed equal averaging on synthetic logistic shards.

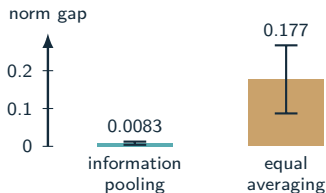
End-to-end trace, four workers

Four concurrent restore, run and capture round trips on islo, timed by the client.

Pooled estimate	4.9422
Full-sample estimate	4.9450
Absolute difference	0.0028
End-to-end trace	6.70 s

Five logistic shards, $n = 60$ to 5,000

Distance to the full-data MLE, lower is better.



Assumes a common parameter and calibrated information. Mean $\pm$ SD over eight fixed seeds, not confidence intervals. Sample-size weighting was not compared.

These checks test aggregation, not agent-task accuracy or dependence correction.

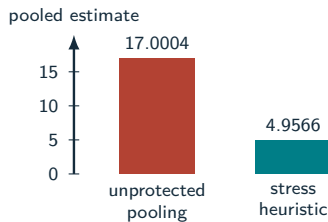
Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4: synthetic checks and trace.

Inflated reported precision pushed the synthetic pooled estimate to 17.0004.

One shard inflates its precision

A 2,000-point shard returns a distant estimate and inflates its reported precision a further 50×.

- Claimed precision raises the shard's weight.
- Sample count alone cannot detect inflated information per point.
- The stress heuristic is not a Byzantine guarantee.



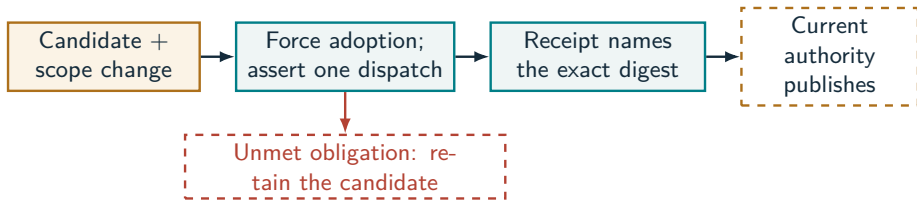
PROPOSED The controller sets each shard's precision by trusted recomputation or calibrated external scoring.

Evidence identity and justified statistical weight are separate requirements.

Synthetic check, Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4. Not an agent-task benchmark.

The proposed gate withholds publication until the missing test passes.

The new same-owner adoption path is a concrete acceptance obligation.



Selection or composition must precede verification of the artifact that will be published.

Export the patch and receipts on every exit, including refusal.

Proposed policy motivated by SELFHOST-2 and SELFHOST-3; not enforced by the numerical reducer.

The proposed cost experiment compares restore with a warm cached template.

	Warm template	Restore
Preparation	Dependencies baked in; pinned source	Capture after setup and one warm test
Fan-out	$N \in \{3, 6, 12\}$; 20 interleaved batches per arm and N	
Endpoint	Batch request $\rightarrow$ last child's first test result	

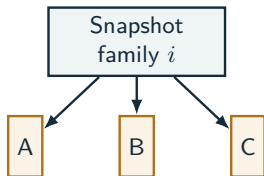
- No model calls; cold start is a reference.
- Establish declared restore fidelity before timing.
- Report capture cost, failed restores and batch latency separately.

A timing win supports this restore path on this workload.

Unrun H1 in PREREGISTRATION.md. Live-memory necessity needs an additional equivalent-state comparison.

The proposed sibling experiment measures environmental co-failure.

Shared parent



Separately prepared parents

Three restores from families $i + 1$, $i + 2$, $i + 3$; matched host and start slot.

12 families; at least 300 lockstep rounds.
Record binary outcome and failure class.

Identify a parent-state feature that capture preserves and the race test consumes.

Unrun H2 in PREREGISTRATION.md; measures environmental outcomes, not hosted-model sampling.

Shared families and repeated rounds require dependence-aware analysis.

- Reusing snapshot families links comparisons.
- Repeated rounds can share scheduling and temporal effects.
- Declare independent blocks or a justified dependence model before collection.

Nine equal-variance measurements	Absolute ρ	N_{eff}
Separately prepared restores	0.10	5.00
Shared-parent restores	0.15	4.09

An excess $\Delta\rho = 0.05$ is insufficient without the baseline dependence.

Illustrative values, not measurements. Proposed amendment; the recorded preregistration remains unchanged.

Testing the disputed path and resampling the repair answer different questions.

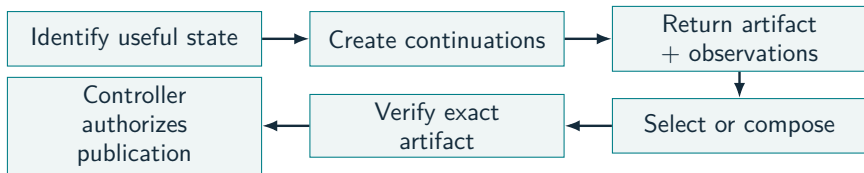
	Targeted behavioral test	Repair resampling
Question	Does this artifact adopt once after same-owner CellBusy?	How repeatable is the out-of-plan edit?
Procedure	Freeze the candidate; force the path; assert one dispatch	Nine samples; fixed input, prompt, policy and hosted endpoint
Record	Executed receipt bound to digest	Scope edits, distinct diffs, claimed test evidence

The targeted test resolves coverage; resampling describes repair behavior.

Both are unrun. Exact equivalence of the public substitute pre-repair tree is unconfirmed.

Proposed regression obligation; descriptive Phase C in PREREGISTRATION.md.

The runtime must preserve shared history and verify the result before publication.



- **State:** exercised paths; useful live-state benefit remains open.
- **Evidence:** duplicate checks built; dependence calibration remains open.
- **Authority:** digest gate built; restore epoch fencing remains open.

H1 chooses the reuse path; H2 tests environmental dependence; the targeted test resolves coverage.

Integrated forked factory and authority protocol remain proposed.